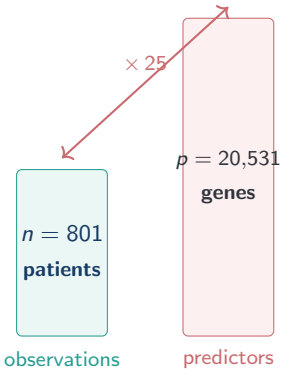
Flexibility vs. Interpretability in High-Dimensional Cancer Classification

Regularized Regression, Tree Ensembles, and Neural Networks on TCGA RNA-Seq Data ($p \gg n$)

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Why Is High-Dimensional Gene Expression Hard?

801 patients, 20,531 measured genes — a severely underdetermined system



Three statistical bottlenecks:

- **Singular covariance:** $\mathrm{rank}(\mathbf{S}) \leq 800 \ll 20{,}531$; OLS and Fisher LDA fail.
- **Separability:** separable data in $p \gg n$ cause MLE divergence ($\|\mathbf{w}\|_2 \to \infty$).
- **Multicollinearity:** shared biological pathways ($|r| > 0.85$) destabilise feature selection.

▷ **Regularization is not optional — it is a mathematical necessity.**

SECTION 01

Dataset and Exploratory Analysis

TCGA RNA-Seq structure, variance screening, and inherent class separability

TCGA RNA-Seq Dataset

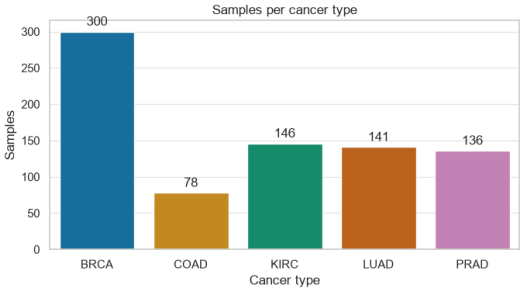
Five primary tumour types — balanced, log-transformed, pre-screened by variance

801 samples
Total patients

1,000 genes
MAD pre-screen

5 classes
BRCA KIRC COAD
LUAD PRAD

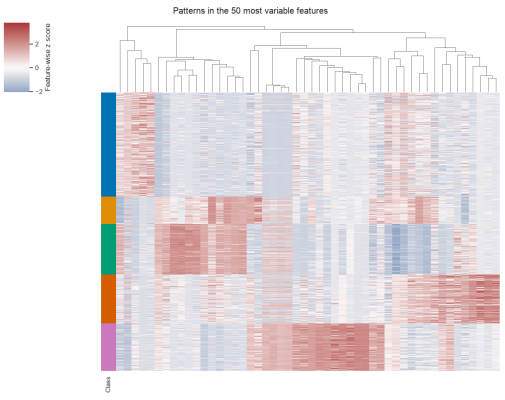
20,531 genes
Raw transcriptome



Preprocessing: Log-transform $z_{ij} = \log_2(x_{ij} + 1)$ ▶ Top-1,000 by in-fold MAD (zero leakage)
▶ In-fold Z-standardization

Expression Heatmap: 50 Highest-Variance Genes

Unsupervised hierarchical clustering reveals tissue-specific expression blocks



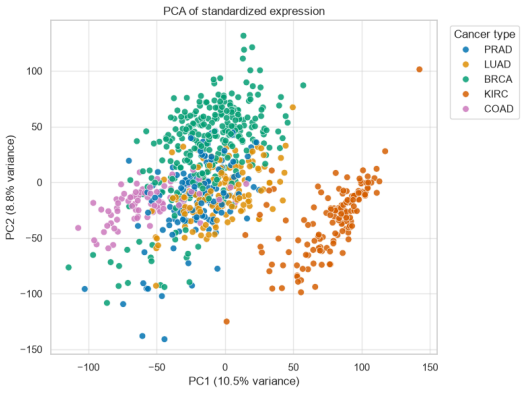
Key observations:

- Colours show standardised expression per gene; columns are patients.
- Cancer cohorts form **distinct blocks** without any labels.
- Row clusters reflect shared **gene modules** and pathways.
- Strong visual separation motivates linear classification.

▷ **The dataset carries strong class-specific structure even in unsupervised clustering.**

PCA: Tumours Separate in Two Linear Dimensions

Spectral projection of 1,000 variance-filtered genes onto the leading principal components



- **PC1 (18.6%):** Separates BRCA from KIRC.
- **PC2 (12.3%):** Separates COAD and LUAD.
- **30.9%** of total variation visible in 2D.
- Five cohorts are almost fully linearly separable in this unsupervised projection.

▷ **Linear separability exists before any model fitting — non-linear methods gain little here.**

SECTION 02

Statistical Methodology

Regularized multinomial regression, tree ensembles, and neural network baselines

Three Regularized Multinomial Classifiers

Same cross-entropy loss $\mathcal{L}_{\text{CE}}$, different penalty geometries under collinearity

Ridge (L_2)

$\mathcal{L}_{\text{CE}} + \frac{\lambda}{2} \sum_k \|\mathbf{w}_k\|_2^2$

- Positive definite Hessian.
- Smooth weight shrinkage.
- Retains **all 1,000 genes**.
- Top calibration (0.005).

Lasso (L_1)

$\mathcal{L}_{\text{CE}} + \lambda \sum_k \|\mathbf{w}_k\|_1$

- Corner solutions (zeros).
- High sparsity: **89 genes**.
- Drops correlated genes arbitrarily.

Elastic Net ($L_1 + L_2$)

$\mathcal{L}_{\text{CE}} + \lambda \sum_k \left[\alpha \|\mathbf{w}_k\|_1 + \frac{1-\alpha}{2} \|\mathbf{w}_k\|_2^2\right]$

- Sparsity + stability.
- Selects **218 genes**.
- **Grouping:** retains co-regulated genes.

Why Elastic Net Handles Collinearity Better

Grouping theorem vs. non-linear baselines

Grouping Bound (Zou & Hastie)

For features with correlation r:

$$\left|\hat{w}_i - \hat{w}_j\right| \leq \frac{\sqrt{2(1-r)}}{\lambda(1-\alpha)} \|\mathbf{y}\|_2$$

- Equal weights as $r \to 1$.
- Correlated genes selected jointly.
- Prevents Lasso's omissions.

Non-Linear Baselines

- **Random Forest:** 200 trees, $m \approx 31$.
- **XGBoost:** 100 rounds, depth 4.
- **PCA-MLP:** 434 PCs (95% var.).
- **Sparse NN:** group-lasso input penalty.

▷ **Trees choose among collinear genes arbitrarily; Elastic Net groups them.**

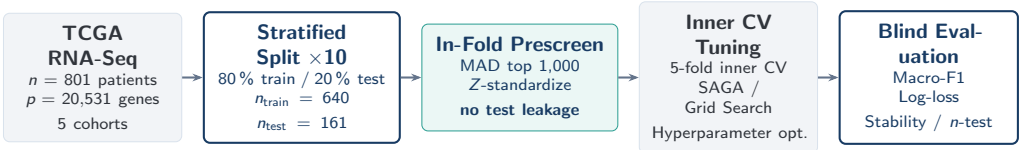
SECTION 03

Experimental Design

Validation protocol, evaluation metrics, and stress tests

Zero-Leakage Evaluation Pipeline

10 repeated stratified splits with strictly within-fold feature prescreening



Strict separation. MAD screening and Z-standardization are computed solely on the training fold. Test data remain entirely blind until final scoring.

Variance estimation. 10 independent random resamplings quantify the variability of every metric, not just its mean.

Why Three Metrics?

Macro-F1 (accuracy), log-loss (calibration), and stability (biomarkers)

Macro-F1

$$\frac{1}{5}\sum_{k=1}^{5}\frac{2P_kR_k}{P_k+R_k}$$

Balanced mean across cohorts.
Robust to cohort size differences.

Limit: all > 0.993; cannot assess calibration.

Log-Loss

$$-\frac{1}{N}\sum_{i,k} y_{ik}\log\hat{p}_{ik}$$

Penalises overconfident errors.
Tests probability quality.

Primary metric for model ranking.

Stability

$$D=\frac{2|S_a\cap S_b|}{|S_a|+|S_b|},\quad J=\frac{|S_a\cap S_b|}{|S_a\cup S_b|}$$

Gene set overlap across resampled training splits.

Key for reproducible biomarkers.

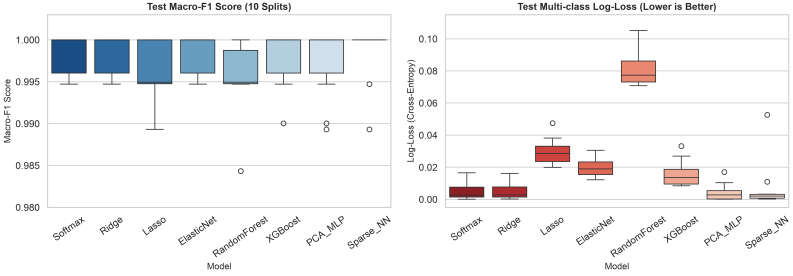
SECTION 04

Results

Benchmark accuracy, calibration, sparsity, and stability across all eight models

All Models Achieve High Accuracy — Log-Loss Tells the Real Story

Performance distributions across 10 repeated train/test splits



F1 saturates above 0.993 for every model. Accuracy alone cannot distinguish architectures on this data.

Log-loss diverges sharply. Random Forest: 0.081. Linear models and PCA-MLP: $<$ 0.020 ($\approx 10\times$ difference).

Comprehensive Model Comparison

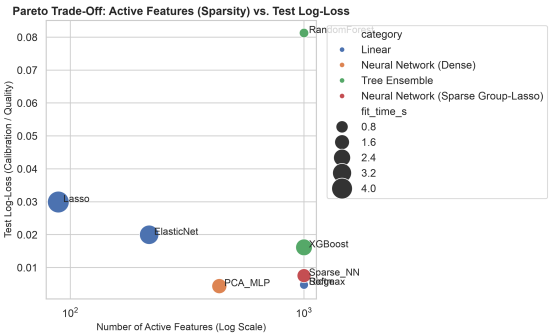
Key metrics: Macro-F1, log-loss, active features, fit time, and selection stability

Model	Macro-F1	Log-Loss	Active Genes	Dice / Jaccard
Softmax (baseline)	0.998 ± 0.003	0.005	1000	1.00 / 1.00
Ridge (L_2)	0.998 ± 0.003	0.005	1000	1.00 / 1.00
Lasso (L_1)	0.995 ± 0.006	0.030	**89**	0.61 / 0.44
Elastic Net	$\mathbf{0.998 \pm 0.003}$	**0.020**	**218**	**0.73 / 0.57**
Random Forest	0.994 ± 0.006	0.081	1000	0.28 / 0.16
XGBoost	0.995 ± 0.006	0.028	1000	0.44 / 0.29
PCA-MLP	0.998 ± 0.003	0.004	434 PCs	—
Sparse NN	0.997 ± 0.004	0.063	1000	—

▷ **Elastic Net matches the best F1, uses only 218 genes, and leads all sparse models in stability.**

Sparsity vs. Calibration: The Pareto Frontier

Fewer active genes means slightly higher log-loss — Elastic Net occupies the efficient frontier

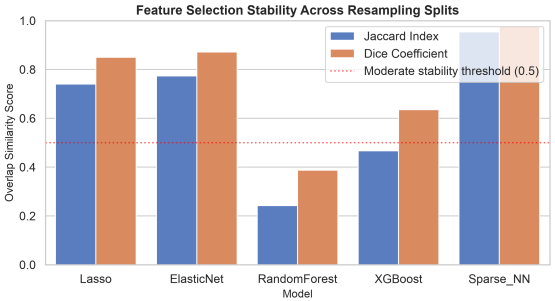


- **Ridge / PCA-MLP:** 1000 genes / 434 PCs, log-loss 0.004–0.005. Dense but calibrated.
- **Lasso:** 89 genes, log-loss 0.030. Highly sparse, slight calibration penalty.
- **Elastic Net:** 218 genes, log-loss 0.020. Sits on optimal frontier.
- **Trees:** dense *and* poorly calibrated (0.081). Off frontier.

▷ **Elastic Net balances biological parsimony and probability calibration.**

Feature Selection Stability

How consistently does each model select the same genes across different data splits?

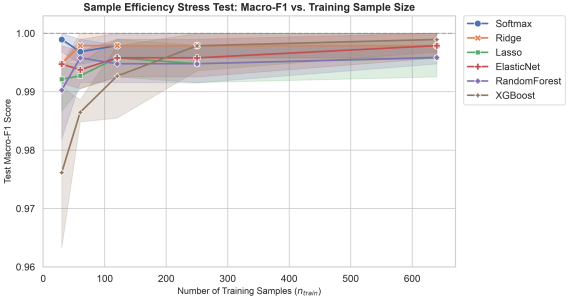


- **Elastic Net:** Dice 0.73, Jaccard 0.57. Highest stability among sparse models.
- **Lasso:** Dice 0.61, Jaccard 0.44. Arbitrary collinear choices hurt stability.
- **XGBoost:** Jaccard 0.29. Greedy splitting is inconsistent.
- **Random Forest:** Jaccard **0.16**. Splits fluctuate with identical impurity gains.

▷ **Biomarker discovery requires L_2 grouping; trees cannot stably rank collinear genes.**

Sample Efficiency: What Happens with Fewer Patients?

Macro-F1 as training set size drops from 640 to 30 (6 patients per class)



- **Linear models** (Ridge, Elastic Net): maintain ≥ 0.995 F1 even down to $n_{\text{train}} = 30$.
- **XGBoost** drops to 0.976 ± 0.018 (min 0.954). Greedy splits overfit noise under scarcity.
- **Random Forest** drops to 0.985 at $n_{\text{train}} = 30$.

Y-axis starts at 0.96 to highlight low-n divergence.

▷ **Regularized linear models remain sample-efficient; trees degrade sharply.**

SECTION 05

Interpretation and Conclusions

Research question answers, limitations, and practical guidance

Three Main Findings

Q1 **Flexibility provides no advantage in RNA-Seq.**

Linear separability holds. Complex models yield no F1 gain and worse probability calibration.

Q2 **Collinearity destabilises Lasso and trees.**

Lasso drops genes arbitrarily; tree stability falls to 0.16. Elastic Net grouping ensures stability (0.57).

Q3 **Elastic Net achieves optimal balance.**

Ties best F1 (0.998), selects 218 genes (78% cut), beats Lasso in calibration (0.020), and leads in stability.

Limitations and Open Questions

Study Limitations:

- **Single cohort:** $n = 801$; batch effects unmeasured across centres.
- **Inter-tissue task:** intra-organ subtypes remain unstudied.
- **Predictive screen:** 218 genes form statistical candidates, not a diagnostic assay.

Non-Linear Roles:

- Subtle intra-tumour molecular subtyping.
- Higher-order epistatic gene interactions.
- Multi-modal data (genomics + pathology).

Takeaway:

Default to Elastic Net for high-dimensional transcriptomics.

In $p \gg n$ RNA-Seq classification,
regularized linear models are not a compromise —
they are the principled choice.

High accuracy, sharp calibration, 78% feature reduction,
and the most reproducible biomarker sets.

Indian Statistical Institute, Kolkata | November 2026

Thank You

Questions & Discussion